BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Random Number Generation

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

BLACK PEN: answer four questions before checking the answers.

1. What does WHERE do in an SQL query?

2. What does a MAC address identify?

A) A website's domain name
B) A network interface
C) A user's password
D) A file's folder

3. "A firewall confirms that every sentence on every website is true." Correct this. Explain why.

4. Data must reach another network in the right order. "TCP chooses fonts; IP draws images." Correct this. Explain each protocol's job.

Today's objectives

Use random values in programs. Use the technique in a different problem.

Key vocabulary - Read, cover, recall

Random numbers
random(low,high) produces a value inside the inclusive range.

Integer range
random(1,6) can return any whole number from 1 to 6.

Unpredictability
Successive random calls can produce different values; outputs are not fixed.

Simulation
Random values can model dice, cards or game events without physical devices.

Write one key term and its meaning in your own words:

Do Now	I Do	We Do	You Do	Review & Improve
Recall previous learning	Read new knowledge	Practise with guidance	Work independently	Check my understanding

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Learn and practise

New learning - Key ideas

Random numbers	random(low,high) produces a value inside the inclusive range. In the dice game, identify where this technique applies.
Integer range	random(1,6) can return any whole number from 1 to 6. In the quiz bonus round, identify where this technique applies.
Unpredictability	Successive random calls can produce different values; outputs are not fixed. In the board-game move, identify where this technique applies.
Simulation	Random values can model dice, cards or game events without physical devices. In the random test selector, identify where this technique applies.

I Do - One worked example

A dice game uses random numbers and integer range. Explain what each contributes. [4 marks]

1. Identify the program's purpose and the pieces of data needed.

2. Random numbers [1]; random(low,high) produces a value inside the inclusive range. [1].

3. Integer range [1]; random(1,6) can return any whole number from 1 to 6. [1].

4. Each feature needs a correct technical fact plus an explanation.

We Do - Guided questions

BLACK PEN: use the model, then explain your answer.

1. A quiz bonus round needs random numbers. Explain the purpose. [2 marks]

2. Explain integer range for a board-game move. [2 marks]

3. When programming a random test selector, explain why unpredictability and simulation matter. [4 marks]

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Write one OCR ERL instruction to generate an integer from 1 to 6 inclusive.

```
roll = ...
print(roll)
```

BLACK PEN: predict or complete the code; test, then improve.

You Do - Independent exam questions

Q2. State TWO relevant ideas or operations from this topic. [2 marks]

Q3. Explain integer range and unpredictability in a virtual spinner. [4 marks]

Q4. A programmer makes a sports draw. Explain the roles of random numbers, unpredictability and simulation. [6 marks]

Exit Ticket - True or False?

Circle T or F before reviewing Page 4.

1. random(low,high) produces a value inside the inclusive range. ☐ T ☐ F
2. random(low,high) produces a value inside the inclusive range ☐ T ☐ F
3. random(1,6) can return any whole number from 1 to 6 ☐ T ☐ F

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: tick correct marking points, then correct errors.

Choose your hardest question to rewrite in full below.

Do Now - Answers

Q1 It selects rows that meet a condition. ☐

Q2 B - A network interface ☐

Q3 It uses rules to allow or block network traffic. ☐

Q4 TCP supports reliable delivery in order. IP handles addressing and routing between networks. Each protocol has its own job; encryption does not prove a site is honest. ☐

Code First - Model

```
roll = random(1,6)
print(roll)
```

Output: Any whole number from 1 to 6 inclusive (not a fixed output)

We Do - Check the model

1. Random numbers [1]; random(low,high) produces a value inside the inclusive range. [1]. ☐

2. Integer range [1]; random(1,6) can return any whole number from 1 to 6. [1]. ☐

3. Unpredictability [1]; Successive random calls can produce different values; outputs are not fixed. [1]. Simulation [1]; Random values can model dice, cards or game events without physical devices. [1]. ☐

Exit Ticket - Check

1. TRUE 2. TRUE 3. TRUE

You Do - Self-mark

Q2 Random numbers [1]; Integer range [1]. ☐

Q3 Integer range [1]; random(1,6) can return any whole number from 1 to 6. [1]. Unpredictability [1]; Successive random calls can produce different values; outputs are not fixed. [1]. ☐

Q4 Random numbers [1]; random(low,high) produces a value inside the inclusive range. [1]. Unpredictability [1]; Successive random calls can produce different values; outputs are not fixed. [1]. Simulation [1]; Random values can model dice, cards or game events without physical devices. [1]. ☐

Try the question again

1. Copy the FULL question you found most difficult:

2. In GREEN PEN, write a complete new answer: